

How to Use Tide

Set it up once, then forget it

USER GUIDE

In Ontario, the same kilowatt-hour costs about 3.9¢ at midnight and 39.1¢ at dinnertime. Tide runs your big appliances at the 3.9¢ hour — automatically, so you never have to think about it.

FOR
Anyone on ULO

UPDATED
May 22, 2026

SETUP
~10 minutes

COMPANION TO
the Team Brief

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Everything you need to set Tide up and live with it — plus the honest limits.

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Set it up once, then forget it. This is the practical companion to `tide-team-brief.md` (which explains why Tide exists) and the spike's `prototypes/tide/README.md` (how to run the code). This page is how a household actually uses it.

The whole point in one line: In Ontario the same kilowatt-hour costs about 3.9¢ at midnight and 39.1¢ at dinnertime. Tide runs your big appliances at the 3.9¢ hour, automatically, so you never have to think about it.

What you'll need

Thing	Notes
A Shelly Plug US Gen4 (~\$25)	The plug Tide switches on and off. We recommend Shelly because it works without internet or a cloud account — control stays inside your home. Any Shelly Gen2+ plug works; cheaper plugs exist from ~\$15.
A device worth shifting	An EV charger (the regular 120-volt kind), a window air conditioner, a dehumidifier, a dishwasher, a pool pump. See What to plug in — and the limits (120-volt loads only).
Somewhere to run Tide	Any always-on computer on your home Wi-Fi — a laptop left plugged in, a Raspberry Pi, a mini PC.
The Ultra-Low Overnight (ULO) rate plan	Call your utility (Toronto Hydro, Alectra, Hydro One...) and ask to switch to ULO. This is what creates the 3.9¢-vs-39.1¢ gap Tide lives in. (A regular Time-of-Use plan also works, with smaller savings.)

Set it up (about 10 minutes, once)

- Wire it up.** Plug the Shelly into the wall. Plug your device (say, your EV charger) into the Shelly. Nothing is shifting yet — that's fine.
- Get the Shelly on your Wi-Fi, then turn Cloud OFF.** In the Shelly's setup screen, connect it to your home network, then switch **Cloud** off. This is the important bit: with Cloud off, Tide talks to the plug directly over your own network, so a flaky internet connection can never break it. Note the plug's address (its IP, like `192.168.1.50`) and, in your router, **reserve that address** so it never changes.
- Point Tide at the plug and test it.** Start Tide and give it the plug's address. Run the connection test — your device should click **on**, hold for two seconds, then click **off**. If the lamp blinks, the hard part is done. (*Today this is one command — see the [appendix](#).*)

4. **Tell Tide about your load — once.** Two facts: **how long it runs** ("my dishwasher takes about 2 hours") and **when it needs to be finished** ("done by 7 a.m."). That's the entire configuration. Tide figures out the rest every night.

Living with Tide

There's no daily routine. That *is* the feature.

- You plug your thing in whenever — 6 p.m., right after work, doesn't matter.
- Every hour, Tide reads two live numbers Ontario publishes for free: **how expensive** power is and **how dirty** it is.
- It finds tonight's cheapest, cleanest stretch that still finishes by your deadline, switches the plug **on** then, and **off** when the job's done.
- You did nothing. You didn't watch the clock. You didn't run the dishwasher at 2 a.m. like a raccoon.

The screen, if you glance at it, shows one word — **WAIT** or **GO NOW** — the live price and carbon, and a running tally of what you've saved. You never *have* to look at it.

What to plug in

The savings, if you shift a load off the 4–9 p.m. window to overnight on ULO:

What you plug in	Saves about
Electric car — full charge	~\$1,760 / year
Electric car — nightly top-up	~\$790 / year
Dehumidifier	~\$320 / year
Window air conditioner	~\$250 / year
Dishwasher	~\$140 / year

Rule of thumb — a load is worth shifting if it (1) runs for a good while, (2) doesn't care *when* it runs, and (3) plugs into a normal wall outlet. The car charger saves the most; a dehumidifier or AC is the most relatable. The \$15–25 plug pays for itself in **days** on an EV, **weeks** on anything else.

Reading your savings

Tide keeps a running ledger — **per run** and **lifetime** — in both dollars and kilograms of CO₂ avoided.

A single EV charge is the clearest example: about **\$2.20** if you'd charged at 6 p.m., versus about **\$0.34** overnight. That's **85% off** — the same energy, the same full battery, a tenth of the price, with zero effort from you.

If something's not working

What you see	The fix
"Can't reach the plug."	Is Cloud turned off in the Shelly app? Is the plug on the same network as the computer running Tide? Is the address right (the one you reserved in step 2)?
The plug switches, but the device doesn't run.	Some appliances don't restart themselves after losing power. Check that yours turns <i>on</i> when power comes back (most chargers, ACs, and pumps do; a few electronics need a physical button).
Flaky or public Wi-Fi (a demo, an apartment, a venue).	Run the plug and the computer on your own little travel router with the reserved address. The grid data + the on-screen display still work even if the plug itself drops off.

What Tide does — and doesn't do

The honest version, so nothing surprises you:

- It **saves money every single day** the prices differ — which on ULO is every weekday.
- It **saves carbon when the hour you shifted *away from* was the dirty one** — true for Ontario weekday evenings, when gas "peaker" plants switch on. On a deep-winter overnight, the *cleaner* benefit shrinks; the *cheaper* benefit never does. So: **money first, planet second**.
- It **shifts *when* you use power, not *how much***. Your total electricity use is the same — Tide just moves it to the cheap, clean hours. It's not an efficiency gadget; it's a timing one.
- **120-volt plug-in loads only**. That covers EV Level-1 charging, window ACs, dehumidifiers, pool pumps. It does **not** cover things hardwired into your panel or on a big 240-volt outlet — electric dryers, Level-2 car chargers, central AC, tank water heaters.
- **You need a time-varying rate**. On a flat rate there's no cheap hour to aim for, so there's nothing to save.

What's real today vs. the finished product

Be clear-eyed about this: the **engine is real and proven**, the **packaging is not built yet**.

What works *right now*, on real Ontario grid data: pulling the live prices and fuel mix, deciding the cheapest-and-cleanest window, switching a real plug on and off and verifying it happened, and tallying the savings. That's the risky part, and it's done.

What's still ahead: turning "run a command and edit one setting" into a one-tap app your aunt could install. The savings don't change — the front door does.

Appendix — running Tide today

For whoever does the technical setup. Full detail in [prototypes/tide/README.md](#).

```
cd prototypes/tide
npm install
npm run plug:test -- 192.168.1.50    # step 3 above — the plug should blink on, then off
npm run plug -- 192.168.1.50        # the real thing: read the grid, switch for the right hour
```

Your load (step 4) is set in `scripts/control-plug.ts` — the `load` object's `durationHours` (how long it runs) and `deadlineHour` (when it must be done). In the finished product this becomes a setup screen.

To see it without any hardware: `npm run sim` (a full 24-hour simulation) or open [prototypes/tide/web/index.html](#) (the WAIT / GO NOW screen).

New to the vocabulary — "the grid," "kWh," "gas peaker"? The mini-glossary at the bottom of [tide-team-brief.md](#) covers every term in one screen.